

# Multi-channel consistency of the matter-bounce prediction at $f_{\text{NL}}^{\text{local}} = -35/16$ : the exact amplitude, pulsar-timing slope, primordial black holes, and survey reach

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(Dated: September 4, 2026)

A dust-dominated contraction before a nonsingular bounce predicts a parameter-free local non-Gaussianity. A from-scratch, vertex-by-vertex in-in computation confirms  $f_{\text{NL}}^{\text{local}} = -35/16 = -2.1875$  (Li *et al.* 2017; Quintin *et al.* 2015), exactly half the value Cai *et al.* (2009) print,  $-35/8$ , locating the discrepancy at Cai *et al.*’s amplitude-normalization step; the squeezed limit is orientation dependent,  $-35/16 + (15/16)\mu^2$ . A classical super-Hubble ( $O(k^0)$ ) route, with non-local shift terms retained, independent of the in-in commutator expansion, reproduces this value exactly. The residual gap to every separate-universe  $\delta N$  value is explained at the equation level: the linear-order relation is a derived identity ( $\delta N_c = (1 - \epsilon/3)\zeta$ ); the second-order piece is a recorded identity whose mechanism is not derived, so the two routes compute different variables rather than disagreeing. Transmission through the bounce is scheme-qualified: within one cubic-vertex scheme (S1) and  $k\eta_B \lesssim 10^{-2}$  – satisfied most easily at large scales, hardest at PBH scales – a linear hand-off bound ( $0 \leq T_{f_{\text{NL}}} < 1/2$ , 0.165–0.409 across three backgrounds) combined with the bounce’s own cubic term gives  $f_{\text{NL}}^{\text{after}} \in [-0.65, -0.50]$  ( $-35/16$ ) or  $[-1.20, -0.86]$  ( $-35/8$ ); a second scheme does not regulate. We take the amplitude through three channels, each stated at the strength its evidence supports. (i) *Pulsar timing*. Refitting NANOGrav’s 15-yr free-spectrum posteriors gives  $\gamma = 2.567 \pm 0.382$ , consistent with the bounce prediction  $\gamma = 3$  at  $1.14\sigma$  (refit) and  $0.55\sigma$  (official posterior,  $\gamma_{\text{HD}} = 3.2_{-0.6}^{+0.6}$ , 5–95%,  $\sigma \approx 0.365$ );  $\gamma = 13/3$  is disfavoured at  $3.1\sigma$ ; insensitive to  $f_{\text{NL}}$ . (ii) *Primordial black holes*. The compaction-function criterion gives a shape-robust required-amplitude ratio  $A(-35/16)/A(-35/8) \approx 1.7$ – $1.9$ , in a regime that is only marginally perturbative ( $1.2|f_{\text{NL}}|\sigma_r \approx 0.5$ – $2$ ) with  $f_{\text{PBH}}(f_{\text{NL}})$  non-monotonic; fed the lab’s predicted spectrum, the channel is a clean *null*: 7.0 orders of magnitude short of the required amplitude, with  $f_{\text{PBH}} = 0$  exactly. (iii) *Large-scale structure*. Consistent with DESI DR1 under either prior ( $0.16\sigma/0.77\sigma$ ). At the transmitted amplitude this channel does not discriminate the two candidates: SPHEREx bare significance is  $0.7$ – $0.9\sigma$  ( $-35/16$ ) or  $1.2$ – $1.7\sigma$  ( $-35/8$ ), under  $1\sigma$  apart. No channel is in tension with another.

## I. INTRODUCTION

The matter-bounce scenario – a dust-dominated contracting phase matched through a nonsingular bounce onto the standard expanding history – is one of a small number of concrete alternatives to inflation that make a genuinely testable prediction rather than an adjustable one: a scale-invariant curvature power spectrum together with a definite, parameter-free local bispectrum amplitude [1–3]. Distinguishing bounce cosmology from inflation, or from  $\Lambda$ CDM with no primordial mechanism at all, is only possible where the two make different, checkable statements; a fixed number with no free parameter, measured in more than one independent channel, is among the strongest such statements available in the near term.

This paper does two things. Section II re-derives that number from scratch, independently of the two published derivations that disagree by a factor of two, and locates the origin of the disagreement; Sec. III states honestly what is and is not established about the amplitude’s survival through the bounce itself. Sections IV–VI then take the resulting value through three independent ob-

servational channels – pulsar timing, primordial black hole abundance, and the large-scale-structure bispectrum – each stated at exactly the strength its evidence supports. Every claim below is graded honestly: a reproduction is labeled reproduced, a first computation is labeled new, and a quoted number is labeled cited and never re-computed silently. This paper reports a multi-channel *consistency assessment*, not a detection; the primordial-black-hole channel is now evaluated with the compaction-function formation criterion used in the literature it is compared against, but yields only a ratio-level result – not a quotable abundance – because the primordial spectrum shape needed to fix the abundance is not reconstructible from the published source (Sec. V A).

## II. THE EXACT MATTER-CONTRACTION AMPLITUDE

Cai *et al.* [1] print a squeezed-limit amplitude  $f_{\text{NL}}^{\text{local}} = -35/8 = -4.375$  for the matter bounce. Li *et al.* [4], whose no-go analysis extends that of Quintin *et al.* [3], obtain exactly one half of it,

$$f_{\text{NL}}^{\text{local}} = -\frac{35}{16} = -2.1875. \quad (1)$$

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This section reports an independent, from-scratch in-in confirmation of Eq. (1), locates the origin of the discrepancy with the printed  $-35/8$ , and states the open item this confirmation does not settle. The full computation, its two validation checks, and the code that produced them are archived and referenced in the reproducibility statement below.

### A. Setup and validation

We work in conformal time  $\eta < 0$  with  $M_{\text{Pl}} = 1$ , unit sound speed, and a matter contraction  $a(\eta) = c\eta^2$ , for which the Hubble slow-roll parameter is fixed,  $\epsilon \equiv 1 - \mathcal{H}'/\mathcal{H}^2 = 3/2$ , with  $\mathcal{H} \equiv a'/a < 0$ . The quadratic action for the comoving-gauge curvature perturbation  $\zeta$  ( $h_{ij} = a^2 e^{2\zeta} \delta_{ij}$ , uniform-field gauge  $\delta\phi = 0$ ) is  $S_2 = \int d\eta d^3x a^2 \epsilon [\zeta'^2 - (\partial\zeta)^2]$ , with Bunch–Davies mode functions fixed by the Wronskian normalization. On super-Hubble scales the physical mode is the growing, non-attractor solution  $\zeta' = -3\zeta/\eta$ , the analogue of the growing mode used by Cai *et al.* [1].

We define the local bispectrum amplitude in the isosceles squeezed configuration  $k_1 \ll k_2 = k_3 \equiv k$  using the same definition as Refs. [1, 4] (the two differ in the evaluation of their shared amplitude normalization below, not in this definition),

$$f_{\text{NL}}^{\text{local}} \equiv \lim_{k_1 \rightarrow 0} \frac{10}{3} \frac{\mathcal{A}(k_1, k, k)}{2k^3}, \quad (2)$$

equivalent to the local template  $\zeta = \zeta_g + \frac{3}{5} f_{\text{NL}} \zeta_g^2$ , using the cubic action derived by Maldacena [5] in comoving gauge for a canonical field: two bulk pieces  $\propto \epsilon^2$ , one boundary/field-redefinition term, and two  $\epsilon^3$  pieces involving  $\tilde{\chi} \equiv \partial^{-2}\zeta'$ . All five are evaluated by first-order in-in perturbation theory, every Wick contraction summed explicitly, every time integral performed exactly before taking  $|k\eta_B| \rightarrow 0$ . Before applying this pipeline to the matter contraction we reproduce two independent literature results with the same code: the de Sitter, constant- $\epsilon$  bispectrum of Ref. [5], matched term by term; and the ultra-slow-roll result of Namjoo, Firouzjahi and Sasaki [6],  $f_{\text{NL}} = 5/2$ . Both match exactly.

### B. Result: a per-vertex table

Table I gives the squeezed-limit contribution of each vertex, from scratch, at  $\epsilon = 3/2$ , in the exact-isosceles configuration  $k_2 = k_3$ , for which  $\mu \equiv \hat{\mathbf{k}}_1 \cdot \hat{\mathbf{k}} = 0$  identically. The total reproduces the  $\mu = 0$  value

$$f_{\text{NL}}^{\text{local}}|_{k_1 \ll k_2 = k_3} = -\frac{35}{16} \quad (3)$$

– this is *not* the angular average  $-15/8$  of Eq. (4) below. Relaxing  $k_2 = k_3$  at  $\mathcal{O}(k_1)$  and defining  $\mu \equiv \hat{\mathbf{k}}_1 \cdot \hat{\mathbf{k}}$  as

TABLE I. Squeezed-limit contribution of each cubic vertex at  $\epsilon = 3/2$ , from a from-scratch in-in computation, to  $f^{\text{sq}}(\mu) \equiv \lim_{k_1 \rightarrow 0} \frac{10}{3} \mathcal{A}(k_1, k, k; \mu)/(2k^3)$ .

| vertex                                           | $f^{\text{sq}}(\mu)$                  |
|--------------------------------------------------|---------------------------------------|
| field redefinition (local + non-local part)      | $-\frac{25}{16} - \frac{15}{16}\mu^2$ |
| $\zeta\zeta'^2$ ( $\epsilon^2 - \epsilon^3/2$ )  | $-\frac{32}{32}$                      |
| $\zeta(\partial\zeta)^2$ <sup>a</sup>            | 0                                     |
| $\zeta'\partial\zeta \cdot \partial\tilde{\chi}$ | $\frac{15}{8}\mu^2$                   |
| $\zeta(\partial_i\partial_j\tilde{\chi})^2$      | $-\frac{15}{32}$                      |
| <b>total</b>                                     | $-\frac{35}{16} + \frac{15}{16}\mu^2$ |

<sup>a</sup> Zero at leading order  $\mathcal{O}(k^2 S^2)$ , where  $S \equiv -\eta_*$  is the (positive) conformal time at which the vertex integral is evaluated, so that  $k^2 S^2 = (k\eta_*)^2 \rightarrow 0$  in the squeezed, super-Hubble limit taken here.

the cosine between the long mode and either (now near-equal) short leg gives, at general orientation,

$$f_{\text{NL}}^{\text{sq}}(\mu) = -\frac{35}{16} + \frac{15}{16}\mu^2, \quad (4)$$

a monopole  $-15/8$  and a quadrupole  $15/16$  under angular averaging. The equilateral and folded values are  $-255/128$  and  $-9/8$ , matching Ref. [4]’s  $c_s = 1$  limit exactly. Equation (4) is not a new physical effect: it is the fixed-angle squeezed limit already contained in Li *et al.*’s Eq. 4.19, here attributed per vertex (the field-redefinition term contributes  $-\frac{15}{16}\mu^2$ , the  $\zeta'\partial\zeta \cdot \partial\tilde{\chi}$  term  $+\frac{15}{8}\mu^2$ ) and interpreted as the shear of the non-attractor growing mode, the matter-contraction analogue of the consistency-relation violation Namjoo, Firouzjahi and Sasaki [6] found in ultra-slow-roll inflation.

### C. The Cai et al. factor of two

Reading the printed shape function of Cai *et al.* [1] (their Eq. 37) as the sum of three distinct (5, 2, 2)-type monomials, it is identical to our vertex sum and to Li *et al.*’s Eq. 4.19 at  $c_s = 1$  (difference zero, checked symbolically). Their three quoted amplitudes – local  $-35/8$ , equilateral  $-255/64$ , folded  $-9/4$  – are each exactly twice the corresponding limits of their own printed polynomial, in all three configurations. We locate the discrepancy between their printed shape function and their quoted amplitudes at this normalization step; this localizes the factor of two to Cai *et al.*’s amplitude-normalization step, specifically their Eqs. (38)–(40), where the printed shape function is converted to the quoted amplitude via  $f_{\text{NL}}^{\text{local}} = (20/3)\mathcal{A}/\Sigma k^3$ . Their printed shape function, Eq. (37), agrees with Li *et al.* [4]’s Eq. 4.19 identically (difference zero, checked symbolically); Li *et al.* redo the in-in calculation for general sound speed and print the amplitude directly in their Eq. 5.1, which reduces to

$-35/16$  at  $c_s = 1$ . Li *et al.* do not themselves comment on the discrepancy with Cai *et al.*'s quoted amplitude; because their shape function (Eq. 4.19) agrees with Cai *et al.*'s (Eq. 37) exactly, the localization above – a uniform factor-2 slip at Cai *et al.*'s amplitude-conversion step – is this paper's own adjudication, not a convention difference reported by either source. Quintin *et al.* [3] likewise quote  $-35/16$  in the text following their cubic-action equation for  $\mathcal{L}_3$ , explicitly attributing it to Cai *et al.* as a citation rather than an independent computation. The present computation is, to our knowledge, the first that reproduces this value from an independent, from-scratch vertex-by-vertex derivation rather than transcribing a prior expression.

#### D. Cross-check by an independent classical route

A separate-universe ( $\delta N$ ) calculation on the same background but evaluated on uniform-density slices gives, for general  $\epsilon$ ,  $f_{\text{NL}}^\rho = 5(\epsilon - 7)/8$ , which reduces at  $\epsilon = 3/2$  to  $(5\epsilon - 35)/8 = -55/16$ , while the comoving-slice  $\delta N$  (an isotropic, shear-free approximation) gives  $f_{\text{NL}}^c = -5$  for all  $\epsilon$ . Neither equals the in-in result above. An anisotropic (Bianchi-I) separate-universe treatment of the long mode's shear response, carrying the projected shear onto the short-mode power spectrum exactly, gives an identically zero monopole contribution: the shear is  $\mathcal{O}(k_L^0)$  and traceless, and no traceless linear-in- $\zeta_L$  response can supply a monopole, so it cannot by itself close the  $-25/8$  gap between  $\delta N^c = -5$  and the in-in monopole  $-15/8$ .

The resolution is that  $\delta N$  on the zero-shift threading computes a different variable from Maldacena's  $\zeta$ . On a uniform- $\phi$  slice the comoving-gauge shift is  $N_i = \partial_i \psi$  with  $\psi \supset a^2 \epsilon \partial^{-2} \zeta$ , which is  $\mathcal{O}(1/k_L)$  in a non-attractor phase – the gradient-expansion assumption  $N_i = \mathcal{O}(\nabla)$  that underlies every separate-universe construction fails here. Separating the long mode's coupling into its local response ([L]: via  $\zeta_L, \dot{\zeta}_L$ ), its shear/curvature ([K]: via  $\partial_i \partial_j \tilde{\chi}_L$ , which cancels exactly between the varied legs of the quartic vertex), its *shift* ([X]: via  $\partial_i \tilde{\chi}_L \propto 1/k_L$ ), and short-mode sourcing ([S], which is suppressed), the classical  $\mathcal{O}(k^0)$  super-Hubble solution reproduces the in-in result of Eq. (4) per vertex and for the full shape function: [X] alone carries 100% of the quadrupole and  $+5/4$  of the monopole, with the  $1/k_L$  poles cancelling only in the sum over vertices – poles no separate-universe construction (isotropic or Bianchi-I) contains. The zero-shift-threading variable and  $\zeta_{\text{Mald}}$  are related at linear order by

$$\delta N_c = \left(1 - \frac{\epsilon}{3}\right) \zeta_{\text{Mald}} \left( = \frac{1}{2} \zeta_{\text{Mald}} \text{ at } \epsilon = \frac{3}{2} \right), \quad (5)$$

derived from the Friedmann equation at fixed  $\phi$  ( $\delta H/H = -(\epsilon/3)\dot{\zeta}_L/H$ ) together with  $\delta N_c = \int (K/3) d\tau$ ; at second order the map is a time-dependent, non-local ( $\partial^{-2}$ ) spatial diffeomorphism, so no local  $f_{\text{NL}}$  relation between  $\delta N_c$

and  $\zeta_{\text{Mald}}$  exists, and comparing their squeezed limits directly (as the naive cross-check does) compares two different observables. The quadrupole gap is likewise resolved rather than merely bounded: the shift's dynamical coupling reproduces the anisotropic-rescaling ( $\propto \mu^2$ ) piece exactly, and a final-time re-threading boundary term – absent from any separate-universe treatment – corrects it to the in-in coefficient exactly.

*Scope statement.*—The from-scratch in-in computation of Table I, independently fixing commutator, Wick-contraction, and orbit multiplicities from scratch and agreeing with two independently published results (Li *et al.*'s  $c_s = 1$  limit and Quintin *et al.*'s quoted value), is the independent second-method route that adjudicates the factor of two *within the in-in method*, and it reproduces  $-35/16$  exactly. The classical  $\mathcal{O}(k^0)$  super-Hubble solution, organized by the shift decomposition above, provides a method-independent confirmation: it reproduces Eq. (4) exactly by a route independent of the in-in commutator expansion. The residual gap to every separate-universe  $\delta N$  value is explained at the equation level rather than reconciled away: at linear order the two calculations are related by the derived identity of Eq. (5); at second order the zero-shift-threading variable  $\delta N_c$  and  $\zeta_{\text{Mald}}$  are related by a time-dependent, non-local spatial diffeomorphism whose net second-order effect on the squeezed bispectrum Appendix A records as an identity, without deriving the mechanism that produces it – so the naive squeezed-limit comparison of the two calculations' raw values compares two different variables, not two disagreeing predictions of the same one. We adopt Eq. (1) throughout this paper as the value of  $\zeta_{\text{Mald}}$ , the variable every downstream channel (transmission, PTA, PBH, LSS) inherits, and carry  $-35/8$  in parallel wherever a channel distinguishes them. Full derivations, the general- $\epsilon$  formula, and the [L]/[K]/[X]/[S] decomposition are given in Appendix A and archived per the reproducibility statement below.

### III. TRANSMISSION THROUGH THE BOUNCE

Equation (4) is a statement about the contraction phase alone, and it holds at every instant of that phase: Sec. II evaluates it in the limit  $|k\eta_B| \rightarrow 0$  and its derivation is end-time independent at leading order (corrections are  $\mathcal{O}(k^2\eta_B^2)$ ). Whether it survives into an observable post-bounce spectrum is a separate, narrower question that depends on physics not computed in full here, and we state the current evidence precisely under one explicit modeling assumption. **(A4):** the contraction-phase bispectrum is frozen at a handoff time  $\eta_h$  near the NEC boundary and evolved purely linearly thereafter – i.e. we bound the mode-mixing *transfer* of the already-fixed contraction-phase amplitude, not the amplitude itself, which Sec. II has already shown is independent of when in the contraction phase it is evaluated. For a linear mode-mixing matrix that carries the growing mode's

amplitude through the bounce with weight 1 and the subdominant decaying mode with weight  $r$ , writing  $\rho(\eta_h) \equiv |J(-\eta_h)|/I_\infty \in (0, 1]$  for the fraction of the mode-mixing integral accumulated before  $\eta_h$ , the transmitted bispectrum amplitude scales as  $T_{f_{\text{NL}}}(\eta_h) = \frac{1+r[1-\rho(\eta_h)]}{1+2r}$ ,

where  $r = -9i\mathcal{A}^2 I_\infty/k^3$  is in general complex; over the validated range  $k\eta_B \lesssim 10^{-2}$  used throughout this paper,  $|r| \gg 1$  holds numerically for all three backgrounds below, and in this growing-mode-dominated limit the transfer reduces to

$$T_{f_{\text{NL}}} = \frac{1-\rho(\eta_h)}{2}, \quad (6)$$

so that, under assumption (A4),  $0 \leq T_{f_{\text{NL}}} < 1/2$  since  $\rho \in (0, 1]$ : within this handoff scheme, linear transfer can only suppress the amplitude, never invert or amplify its sign. Evaluated across three explicit nonsingular bounce backgrounds – an LQC effective quasi-dust bounce (“LQC”), an analytic non-LQC polynomial bounce (“poly (analytic non-LQC)”), a different functional bounce profile from the LQC effective-dust form, not an LQC sub-case), and the Quintin *et al.* [3]-type parametrization (“Quintin-type”) – at the natural handoff  $\eta_h = \eta_B$  (the NEC boundary),  $T_{f_{\text{NL}}}$  ranges over 0.165–0.409: a  $1.64\times$  spread between two mode-function conventions (geometric vs. effective-fluid) on the *same* LQC background (0.409/0.250), and a  $2.48\times$  spread across the full set of three backgrounds and two conventions (0.409/0.165). This bound is a statement about scheme-dependent linear transfer, not a universal property of the physical post-bounce  $f_{\text{NL}}$ .

*Direction of the  $k\eta_B$  window.*—The validity condition  $k\eta_B \lesssim 10^{-2}$  used throughout this section is an *upper* bound on  $k$ , not a lower one: it is satisfied *most easily* at the largest observable scales (the CMB/LSS pivot,  $k \sim 0.05 \text{ Mpc}^{-1}$ ) and *hardest* to satisfy at the smallest, PBH-relevant scales ( $k \sim 10^{15}\text{--}10^{16} \text{ Mpc}^{-1}$  for a  $10^{15} \text{ g}$  PBH), where it requires a correspondingly high bounce energy (Sec. VC). Consequently the scheme-S1 transfer of this section is satisfied most comfortably at the LSS/CMB pivot used by Channel III below, and the transmitted amplitude  $f_{\text{NL}}^{\text{after}} = T_{f_{\text{NL}}} \cdot f_{\text{NL}}^{\text{before}} + \Delta f_{\text{NL}}^{\text{bounce}}$  (Sec. III A) is therefore this paper’s observable prediction for that channel, not the pre-bounce value.

What this bound does not include is the bounce’s own cubic self-interaction term, which the loop-quantum-cosmology literature finds can *enhance* non-Gaussianity by orders of magnitude through gravitational self-interactions in the third-order Hamiltonian, even in the absence of a potential [7]; its sign relative to the contraction-phase value in Eq. (1) is not established by the present work and is cited here, not computed. Any assumption that the bounce transmits  $f_{\text{NL}}$  with a coefficient near unity is therefore not supported by either calculation: the linear channel caps it at  $1/2$  within the (A4) handoff scheme.

## A. The bounce’s own cubic term: now computed

The intrinsic bounce contribution left open above is closed for one explicit scheme. In the geometric (dressed-metric) prescription  $z = a$ ,  $\epsilon_{\text{eff}} = 1/2$ ,  $c_s = 1$  – scheme S1, the substitution of these values into the classical comoving-gauge  $P(X, \phi)$  cubic action of Ref. [5], an assumption of the scheme and not a derivation of the quantum-gravity-dressed operator – the squeezed-isocles bounce contribution evaluated over the window  $[-\eta_B, \eta_B]$  is, with  $\rho_B \equiv \rho(\eta_B) = 1 - 2T_{f_{\text{NL}}}$  the same handoff fraction of Sec. III evaluated at the bounce,

$$\Delta f_{\text{NL}}^{\text{bounce}}[\text{S1}] = -\frac{5}{24}\rho_B, \quad (7)$$

so that, combined with the linear transmission of the contraction-phase amplitude below,  $f_{\text{NL}}^{\text{after}} = -\frac{85}{48}T_{f_{\text{NL}}} - \frac{5}{24}$ . a closed-form vertex-table result confirmed to  $3 \times 10^{-4}$  by an independent finite- $k$  numerical in-in evaluation of all six cubic attachments and boundary terms; the vertex  $\zeta\dot{\zeta}^2$  carries 99.95–99.97% of it. Combining Eq. (7) with the linear transmission of Eq. (1) at  $\eta_h = \eta_B$ ,  $f_{\text{NL}}^{\text{after}} = T_{f_{\text{NL}}} \cdot (-35/16) + \Delta f_{\text{NL}}^{\text{bounce}}$ , gives a definite post-bounce prediction for the first time, one number per background:

| background              | $T_{f_{\text{NL}}}$ | $\Delta f_{\text{NL}}^{\text{bounce}}$ | $f_{\text{NL}}^{\text{after}}$ |
|-------------------------|---------------------|----------------------------------------|--------------------------------|
| Quintin-type [3]        | 0.165               | −0.140                                 | −0.501                         |
| LQC effective dust      | 0.250               | −0.104                                 | −0.651                         |
| poly (analytic non-LQC) | 0.196               | −0.127                                 | −0.555                         |

so  $|f_{\text{NL}}^{\text{after}}| \in [0.50, 0.65]$  against  $|f_{\text{NL}}^{\text{before}}| = 2.19$ : the bounce’s own term is negative – it enhances  $|f_{\text{NL}}|$ , partially undoing but not reversing the linear suppression, at 19–39% of the transmitted contraction term. This range is valid only for  $k\eta_B \lesssim 10^{-2}$  (set by the competing requirements that the post-bounce integral converge and the super-Hubble boundary treatment hold) and only within scheme S1; a second scheme (effective-fluid, S2) gives a divergent  $\Delta f_{\text{NL}}^{\text{bounce}}$  with no regulated value ( $d_{\text{cut}}^{-1}$  scaling as  $d_{\text{cut}} \rightarrow 0$ , the cubic-order restatement of the linear-order  $z^2$  pathology) and is reported as a non-result, not quoted numerically anywhere in this paper. Quintin *et al.*’s reported bispectrum growth through a nonsingular bounce [3] is not contradicted: their growth is the *additive* bounce-generated term, the counterpart of Eq. (7), sharing its sign (both enhance  $|f_{\text{NL}}|$ ); the factor-of-two suppression identified above is the *multiplicative* transmission of the inherited contraction amplitude, a quantity they do not compute, so the two statements are about different terms and are not in tension. Agullo, Bolliet & Sreenath’s LQC enhancement [7] remains not comparable as computed: scheme S1 contains the classical counterpart of their dominant operator but not its quantum-geometric dressing, so the present result neither reproduces nor refutes theirs. The



nonlinear channel is therefore no longer an unknown but a scheme-limited, qualified prediction; every use of  $f_{\text{NL}}^{\text{after}} \in [-0.65, -0.50]$  in this paper carries the “scheme S1” qualifier, and the unqualified statement that the bounce produces no orders-of-magnitude enhancement is not established. The full vertex table, regularization-assumption note, numerical cross-check, and literature-comparison derivation are archived and referenced in the reproducibility statement below.

#### IV. CHANNEL I: THE PULSAR-TIMING SPECTRAL SLOPE

##### A. Spectral conventions

With the characteristic strain parameterised as  $h_c(f) \propto f^{(3-\gamma)/2}$  and  $\Omega_{\text{GW}} = (2\pi^2/3H_0^2) f^2 h_c^2(f)$ , the energy density scales as

$$\Omega_{\text{GW}}(f) \propto f^{5-\gamma}. \quad (8)$$

Hence  $\gamma = 13/3$  for gravitational-wave-driven supermassive-black-hole binary inspirals ( $\Omega_{\text{GW}} \propto f^{2/3}$ ),  $\gamma = 5$  for a scale-invariant primordial tensor background ( $n_{\text{T}} = 0$ ,  $\Omega_{\text{GW}} \propto f^0$ ), and  $\gamma = 3$  for  $\Omega_{\text{GW}} \propto f^2$ , the universal infrared scaling of the scalar-induced background of a nonsingular matter bounce [8]. We stress that  $\gamma = 3$  is *not* the  $n_{\text{T}} = 0$  case; the two are separated by two units of slope.

##### B. Official posterior and this lab’s refit

The primary comparison for this channel is NANOGrav’s own published result: fitting the 15-yr Hellings–Downs-correlated free-spectrum posterior on the lowest 14 frequency bins with a power law, NANOGrav report  $\gamma_{\text{HD}} = 3.2^{+0.6}_{-0.6}$ , quoted explicitly as a posterior median and 5–95% interval [10]; a Gaussian-equivalent  $1\sigma \approx 0.6/1.645 \approx 0.365$ . Against this official posterior,  $\gamma = 3$  (matter-bounce,  $\Omega_{\text{GW}} \propto f^2$ ) sits at  $0.55\sigma$ ,  $\gamma = 5$  (scale-invariant primordial tensors,  $n_{\text{T}} = 0$ ) at  $4.94\sigma$ , and  $\gamma = 13/3$  (SMBHB) at  $3.1\sigma$  – consistent with NANOGrav’s own characterization of  $13/3$  as in “moderate tension ... at the 99% credible boundary.”

As a secondary, differently-conditioned cross-check, we independently refit the same Ceffyl kernel-density free-spectrum posteriors using *all* 30 frequency bins [10, 11] ( $T_{\text{obs}} = 16.03 \text{ yr}$ ) with a power law ( $\gamma, \log_{10} A$ ) under priors  $\gamma \sim U[0, 7]$ ,  $\log_{10} A \sim U[-18, -11]$ , using 32 walkers and  $10^4$  production steps after 2500 burn-in ( $3.2 \times 10^5$  samples; integrated autocorrelation time  $\tau = (58.1, 58.0)$  for  $(\gamma, \log_{10} A)$ , giving  $\text{ESS} = N/\max(\tau) = 5.5 \times 10^3$ ). The marginal is

$$\gamma = 2.567 \pm 0.382, \quad \log_{10} A = -14.03 \pm 0.38, \quad (9)$$

with median 2.591 and 68% interval  $[2.304, 2.882]$ ; this refit pipeline is validated by injection (Sec. IV C) and sits

TABLE II. Channel I. Tensions against the official NANOGrav 14-bin posterior ( $\gamma_{\text{HD}} = 3.2^{+0.6}_{-0.6}$ ,  $\sigma \approx 0.365$ ) and against this lab’s 30-bin refit (Eq. (9)), with the refit’s Savage–Dickey factor where the chain has posterior support. All values reproduced from the committed chain (`pta_gamma_reproduce.py`); differences from a self-reproduction run of the same script against the same chain are  $\leq 3 \times 10^{-15}$  (floating-point-level agreement, not an independent external check).  $z_{\text{refit}}$  at  $\gamma = 5$  is a Gaussian approximation only: 0 of  $3.2 \times 10^5$  chain samples reach  $\gamma \geq 5$  (chain max 4.70), so no data-supported Bayes factor is reported there (see text).

| Hyp.          | $\gamma_*$ | slope     | $z_{\text{off}}$ | $z_{\text{refit}}$ | $B_{\gamma_*/\text{free}}$ |
|---------------|------------|-----------|------------------|--------------------|----------------------------|
| Matter bounce | 3          | $f^2$     | $0.55\sigma$     | $1.14\sigma$       | 3.23                       |
| SMBHB         | 13/3       | $f^{2/3}$ | $3.1\sigma$      | $4.63\sigma$       | $5 \times 10^{-4}$         |
| Prim. tensors | 5          | $f^0$     | $4.9\sigma$      | $6.37\sigma$       | n/a                        |

$1.20\sigma$  from the official posterior (quadrature-combined  $\sigma = 0.53$  from the two marginals’ widths). Table II reports both the official-posterior tensions and Gaussian  $z$ -distances against the refit’s own marginal, plus Savage–Dickey density ratios against the refit,  $B_{\gamma_*/\text{free}} = p(\gamma_*|d)/\pi(\gamma_*)$  with  $\pi = 1/7$ , quoted only where the chain has posterior support: with zero of  $3.2 \times 10^5$  samples at  $\gamma \geq 5$  (chain maximum 4.70), no Bayes factor at  $\gamma = 5$  is reported – a KDE evaluated there is extrapolating into an unsampled tail, and its value moves by an order of magnitude under a different bandwidth choice. The  $\gamma = 13/3$  factor rests on 9 samples at  $\gamma \geq 13/3$  and is quoted to one significant figure for the same reason.

The  $1.14\sigma$  and  $4.63\sigma$  refit entries are Gaussian approximations to a marginal that is not itself Gaussian (it is bounded by the sampling prior); directly from the chain,  $P(\gamma > 3) = 8.97\%$ . Narrowing the refit’s  $\gamma$  prior from  $U[0, 7]$  to  $U[2, 5]$  moves  $B_{3/\text{free}}$  from 3.23 to 1.47 while leaving  $B_{\text{MB}/\text{SMBHB}} = B_{3/\text{free}}/B_{13/3/\text{free}} \approx 7 \times 10^3$  ( $\log_{10} B \approx +3.9 \pm 0.2$ , one significant figure given the 9-sample tail estimate) stable in the range  $7\text{--}9 \times 10^3$ ; the individual Savage–Dickey factors  $B_{\gamma_*/\text{free}}$  against the refit’s own prior (“nested” since each  $\gamma_*$  is a point restriction within the refit’s free- $\gamma$  model) are prior-sensitive, but their ratio  $B_{\text{MB}/\text{SMBHB}}$  (the model ratio) is not.

##### C. Refit validation

The refit’s likelihood/prior construction (30 bins,  $T_{\text{obs}} = 16.03 \text{ yr}$ ,  $\gamma \sim U[0, 7]$ ,  $\log_{10} A \sim U[-18, -11]$ ) is validated by an injection-recovery test that reuses that construction verbatim, using the REAL NANOGrav 15-yr HD-correlated free-spectrum KDE density grids from the public Zenodo record [10.5281/zenodo.8060824](https://zenodo.org/record/8060824/files/30f_fs{hd}_ceffyl/*.npy) (30f\_fs{hd}\_ceffyl/\*.npy, sha256-verified). Each bin’s real observed KDE curve (shape/width/skew preserved exactly) is re-centered along the  $\log_{10} \rho$  axis to the model prediction at a chosen injected ( $\gamma_{\text{true}}, \log_{10} A_{\text{true}}$ ), since a real dataset’s true signal is unknown and a

ground-truth injection test requires a known target; this is the strongest injection achievable from real data without simulating a full new PTA dataset. At  $\gamma_{\text{true}} = 13/3$ , five realizations (the full 30-bin recovery plus four bootstrap bin-resamples) recover  $\gamma = 4.336$  on average with a mean pull of  $+0.016 \pm 0.044\sigma$  (std 0.0993 over the 5 realizations, standard error of the mean); at a  $\gamma_{\text{true}} = 3$  control, five realizations recover  $\gamma = 3.005$  with a mean pull of  $+0.033 \pm 0.054\sigma$  (std 0.1203; script, JSON, and data manifest: `research/track_a3_multichannel/pta_injection_30bin_realkde_2026_09_02.py`, `outputs/pta_injection_30bin_realkde_2026_09_02.json`). Both are consistent with unbiased recovery: both mean pulls are well under  $0.1\sigma$ , though the per-realization scatter (std  $\approx 0.10$ – $0.12\sigma$  over 5 realizations) is comparable to, not smaller than, that threshold, tightening a secondary synthetic-density cross-check that substitutes a synthetic per-bin Gaussian density for the real KDE grids:  $\gamma_{\text{true}} = 13/3 \rightarrow$  mean pull  $-0.026\sigma$ ;  $\gamma_{\text{true}} = 3 \rightarrow$  mean pull  $+0.068\sigma$  (`pta_injection_30bin_2026_09_02.py`) – both runs agree the pipeline is unbiased at well under  $0.1\sigma$ .

#### D. What this channel does and does not establish

Both the official posterior and the refit are consistent with the matter-bounce induced-GW prediction  $\gamma = 3$  and disfavour both the SMBHB expectation and a scale-invariant primordial tensor background, though the strength of that disfavouring differs by conditioning:  $3.1\sigma$  against the official posterior and  $4.63\sigma$  against this lab’s 30-bin refit (Table II); the latter should not be read as tighter than NANOGrav’s own official assessment, since it conditions on a different (refit) marginal. This channel does not favour the matter bounce *uniquely* on causality grounds alone: the universal causality-limited infrared tail of an induced background sourced by a finite-duration process is  $\Omega_{\text{GW}} \propto f^3$  ( $f^3 \ln^2 f$  for a narrow spectral peak), *not*  $f^2$  [9], so  $\Omega_{\text{GW}} \propto f^2$  is *shallower* than that causal floor and is not generic to scalar-induced backgrounds of arbitrary origin. Here  $\gamma = 3$  ( $\Omega_{\text{GW}} \propto f^2$ ) instead follows from the matter bounce’s broad, near-scale-invariant scalar source:  $\Omega_{\text{GW}} \propto P_{\mathcal{R}}^2$  (Papanikolaou’s Eq. (30)) combined with the source spectrum’s own near-linear low- $k$  growth  $P_{\mathcal{R}} \propto k$  (Eq. (8) there) gives  $\Omega_{\text{GW}} \propto k^2 \propto f^2$  [8]; the cited derivation does not state whether this holds only asymptotically at the lowest frequencies or across the full NANOGrav band, so we do not claim more than the IR asymptotic scaling here.  $\gamma = 3$  is therefore *more* distinctive than a generic causality argument would suggest, not less: it requires the specific broad, near-scale-invariant source spectrum of the matter bounce, not merely any finite-duration scalar-induced background. Nor does this channel constrain  $f_{\text{NL}}$ : it measures a slope, not a bispectrum amplitude, and is identical at  $-35/16$  and  $-35/8$ . Converting it into a dis-

criminating test requires propagating the matter-bounce scalar spectrum through the induced-GW kernel to predict an *amplitude* in the nHz band; that computation is not attempted here.

## V. CHANNEL II: PRIMORDIAL BLACK HOLE ABUNDANCE

### A. Press–Schechter with local quadratic non-Gaussianity: a truncated-map bound

Writing  $\zeta = \zeta_G + A(\zeta_G^2 - \sigma^2)$  with  $A = \frac{3}{5}f_{\text{NL}}$  and  $\zeta_G \sim \mathcal{N}(0, \sigma^2)$  [12, 13], the mass fraction is  $\beta(\zeta_c) = P(\zeta > \zeta_c)$ , obtained exactly by inverting the quadratic. For  $f_{\text{NL}} < 0$  this map is a downward parabola with an absolute ceiling

$$\zeta_{\text{max}} = -\frac{5}{12f_{\text{NL}}} + \frac{3}{5}|f_{\text{NL}}|\sigma^2, \quad (10)$$

whose *leading* term scales as  $1/|f_{\text{NL}}|$  and exactly doubles when  $|f_{\text{NL}}|$  is halved; quoting that leading term alone gives  $\zeta_{\text{max}} = 0.09524$  at  $-35/8$  and  $0.19048$  at  $-35/16$ . The subleading  $\frac{3}{5}|f_{\text{NL}}|\sigma^2$  term is dropped from those two numbers; restoring it at a representative  $\sigma = 0.1$  gives the full  $\zeta_{\text{max}} = 0.1215$  at  $-35/8$  and  $0.2036$  at  $-35/16$  (ratio 1.68, not exactly 2). At the standard curvature thresholds  $\zeta_c \approx 0.45$ – $1$  this ceiling forces  $\beta \equiv 0$  identically for *both* candidate values, so a first pass restricted to the low-threshold regime ( $\zeta_c \lesssim 0.15$ ) is the only place the quadratic map produces a nonzero answer at all – and there it reported  $f_{\text{PBH}}(-35/16) > f_{\text{PBH}}(-35/8)$  by three to ten orders of magnitude, with the halved value landing inside the  $10^{-3} \leq f_{\text{PBH}} \leq 1$  band reported by Choudhury *et al.* [15] for negative- $f_{\text{NL}}$  bounce scenarios. That ordering was the sharpest statement the channel could make, but the ceiling that produced it is an artefact of truncating the local map at quadratic order, not a physical statement: it is exactly why the literature this comparison is drawn against uses the compaction-function formation criterion instead [15], and redoing the calculation on that criterion is the subject of the rest of this section.

### B. The compaction-function result

Following Young, Byrnes & Sasaki (2019) [16] and Choudhury *et al.* (2025) [15], formation is set by the peak compaction  $C(r_p) = \delta_p$  exceeding a threshold  $C_{\text{th}}$ , with the radiation-era compaction function  $C = C_{\text{lin}} - C_{\text{lin}}^2/(4f(w))$ ,  $f(w) = 2/3$ , and  $C_{\text{lin}} = JC_G$  carrying the same quadratic local map ( $J = 1 + \frac{6}{5}f_{\text{NL}}\zeta_G$ ) applied to the linear compaction  $C_G$  rather than to  $\zeta$  itself. Because  $C_G$  is an unbounded Gaussian variable, the map  $\zeta_G \rightarrow C_{\text{lin}}$  has *no* ceiling, so every threshold  $C_{\text{th}} < C_{\text{max}} = f(w) = 2/3$  is reachable at every  $f_{\text{NL}}$ : the Eq. (10) artefact is removed by construction. We integrate the joint Gaussian PDF of  $(C_G, \zeta_G)$

over the type-I formation domain (Musco 2019 [17];  $C_{\text{th}}$  scanned over  $\{0.4, 0.5, 0.6\}$ ) with a critical-scaling mass function ( $\gamma_{\text{crit}} = 0.36$ ,  $K = 4$ ), using the Gaussian window for  $C_G$  and the spherical window (with radiation transfer function) for  $\zeta_G$ , exactly as in Choudhury *et al.* Eqs. (30–66). The one ingredient of their calculation that is not reconstructible from the published paper is the regularized-renormalized- resummed one-loop curvature spectrum sourced by their EFT-of-bounce-plus-USR mode history: no closed-form  $\Delta_\zeta^2(k)$ , loop-counterterm normalisation, or EFT parameter set is printed. We use a lognormal stand-in spectrum, scanned over width  $\Delta \in \{0.35, 0.5, 0.8\}$  and peak-scale  $r_p k_p \in \{0.75, 1.0, 1.5\}$ , and scan the amplitude  $A$  rather than adopting theirs.

At fixed curvature amplitude, calibrated so the Gaussian case gives  $f_{\text{PBH}} = 1$ , this criterion *reverses* the first-pass ordering: across the full 27-point  $(\Delta, r_p k_p, C_{\text{th}})$  grid,  $f_{\text{PBH}}(-35/16) < f_{\text{PBH}}(-35/8)$  at *every* point (Table III, Fig. 1), because the correlation  $\gamma_{\text{cr}} \equiv \sigma_{\text{cr}}^2 / (\sigma_c \sigma_r)$  – with  $\sigma_c^2$  the compaction-function variance,  $\sigma_r^2$  the curvature variance, and  $\sigma_{\text{cr}}^2$  their cross-variance, all evaluated with the same real-space top-hat window at the peak scale  $r_p$  used for  $C_{\text{th}}$  – that governs the sign of the non-Gaussian correction is large and positive (0.766–0.968) at these spectrum shapes: the Gaussian-case formation channel rides a correlated ridge that negative  $f_{\text{NL}}$  destroys, forcing the system into an anti-correlated, probability-costly regime whose leverage *grows* with  $|f_{\text{NL}}|$ .  $-35/8$  therefore always out-produces  $-35/16$  here – the opposite of the truncated-quadratic-map result. The one analytic statement of the first pass, that Eq. (10)’s leading term exactly doubles when  $|f_{\text{NL}}|$  is halved (the full expression, including the dropped  $\sigma^2$  term, does not), remains arithmetically true of that leading term but is now physically empty: it described a truncation artefact that the compaction criterion does not share.

$f_{\text{PBH}}$  itself is not a quotable constraint: because it depends exponentially on  $\gamma_{\text{cr}}$ , it moves by more than 100 decades across the grid at fixed target amplitude (Table III), tracking the unreproducible spectrum shape (the primordial spectrum’s unreconstructible detailed profile, discussed further in Sec. VB). What *is* robust is the ratio of curvature amplitudes needed to reach a fixed target abundance, since  $K$ ,  $\Omega_{\text{DM}}$ ,  $g_*$ , and  $M_H$  all cancel in it: reaching the floor of the Choudhury *et al.* band,  $f_{\text{PBH}} = 10^{-3}$ , requires

$$\frac{A(-35/16)}{A(-35/8)} = 1.732 [1.610, 1.809], \quad \text{std} = 0.050, \quad n = 27, \quad (11)$$

i.e. stable to  $\pm 6\%$  while  $f_{\text{PBH}}$  itself moves by more than 100 dex. This ratio is the channel’s most reproducible output – it holds at every one of the 27 grid points and is robust to spectrum shape (std = 0.050 on a mean of 1.732) – and it is a result within a stated regime, not a claim of general validity. *Regime of validity.* The result lives on the anti-correlated ( $\gamma_{\text{cr}} > 0$ ,  $J > 1$ ) branch identified above, and it holds at curvature excursions where the quadratic local map is not always perturbatively con-

TABLE III. Channel II. Compaction-function criterion: amplitude ratio  $A(-35/16)/A(-35/8)$  required to reach  $f_{\text{PBH}} = 10^{-3}$ , and the *nominal (uncapped)*  $f_{\text{PBH}}$  columns – i.e. evaluated at the Gaussian-calibrated amplitude  $A_*$  with no upper cutoff applied – for representative points of the  $(\Delta, r_p k_p, C_{\text{th}})$  grid (full 27-point grid in `outputs/pbh_compaction_fnl.json`).

| $\Delta$                                             | $r_p k_p$ | $C_{\text{th}}$ | $\gamma_{\text{cr}}$ | $f_{\text{PBH}}(-35/16)$ | $f_{\text{PBH}}(-35/8)$ | ratio $A$ |
|------------------------------------------------------|-----------|-----------------|----------------------|--------------------------|-------------------------|-----------|
| 0.35                                                 | 0.75      | 0.5             | 0.897                | $5.8 \times 10^{-21}$    | $4.5 \times 10^{-7}$    | 1.767     |
| 0.35                                                 | 1.5       | 0.5             | 0.968                | $5.3 \times 10^{-107}$   | $7.0 \times 10^{-59}$   | 1.673     |
| 0.50                                                 | 1.0       | 0.5             | 0.888                | $3.6 \times 10^{-14}$    | $1.6 \times 10^{-2}$    | 1.734     |
| 0.80                                                 | 0.75      | 0.6             | 0.766                | $3.5 \times 10^3$        | $2.2 \times 10^8$       | 1.809     |
| 0.80                                                 | 1.5       | 0.4             | 0.857                | $5.8 \times 10^{-9}$     | $1.6 \times 10^1$       | 1.677     |
| mean ratio 1.732 [1.610, 1.809], std 0.050, $n = 27$ |           |                 |                      |                          |                         |           |

trolled:  $1.2|f_{\text{NL}}|\sigma_r \approx 0.54$ – $1.01$  at  $-35/16$  and  $\approx 1.09$ – $2.02$  at  $-35/8$  across the grid – the same class of limitation Choudhury *et al.* report as their own  $|f_{\text{NL}}| \lesssim 60$  perturbativity bound, and a limitation of the shared quadratic ansatz rather than of either calculation specifically. *Non-monotonicity.* At fixed curvature amplitude,  $f_{\text{PBH}}(f_{\text{NL}})$  is strongly non-monotonic along the continuity path from the Gaussian case ( $f_{\text{NL}} = 0$ ) to  $-35/8$ : it falls by  $\sim 55$  decades to a minimum near  $f_{\text{NL}} \approx -0.35$ , then climbs back by  $\sim 53$  decades through  $-35/16$  to  $-35/8$ . Both candidate values lie past that minimum, on the branch where the anti-correlated ( $\zeta_G < 0$ ) region of  $(C_G, \zeta_G)$  space dominates formation and drives the ratio’s leverage; a reader inspecting only the two candidate points would not see this structure, so we record it here for transparency even though it does not change Eq. (11). The Gaussian-calibrated normalization amplitude at the representative grid point  $(\Delta, r_p k_p, C_{\text{th}}) = (0.5, 1.0, 0.5)$  is  $A_* = 0.131446$  (the amplitude at which  $f_{\text{PBH}} = 1$  for  $f_{\text{NL}} = 0$ ); Table III and Fig. 1 report  $f_{\text{PBH}}$  at this calibration. Reproducing Choudhury *et al.*’s reported band at  $-35/8$  ( $10^{-3} \leq f_{\text{PBH}} \leq 1$ ) is not forced by construction – the amplitude was fixed by the unrelated Gaussian-case normalisation – and it lands there unforced for  $\gamma_{\text{cr}} \gtrsim 0.85$ ; for  $\gamma_{\text{cr}} \lesssim 0.85$ , at the same fiducial  $f_{\text{NL}} = -35/8$  Choudhury *et al.* report, our implementation instead finds *enhancement* relative to Gaussian where they report suppression, a genuine discrepancy left unresolved because it depends on which side of that crossover their (unreproducible) spectrum falls.

Present-day abundance uses [14]

$$f_{\text{PBH}} = \frac{1}{\Omega_{\text{DM}}} \left( \frac{M_\odot}{M_H} \right)^{1/2} \left( \frac{g_*}{106.75} \right)^{3/4} \left( \frac{g_{*s}}{106.75} \right)^{-1} \times \frac{\beta_{\text{NG}}}{7.9 \times 10^{-10}}, \quad (12)$$

with  $\Omega_{\text{DM}} = 0.674^1$ ,  $g_* = g_{*s} = 106.75$ , and a sin-

<sup>1</sup> Quoted as printed in Choudhury *et al.*’s Eq. (66); it numerically

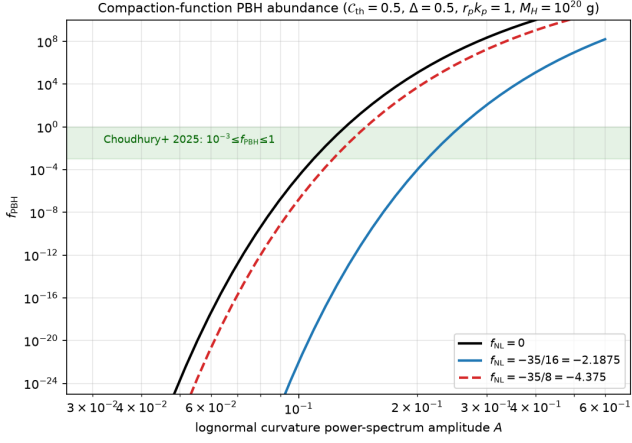


FIG. 1. Channel II, compaction-function criterion:  $f_{\text{PBH}}$  vs. curvature power-spectrum amplitude  $A$  at the representative grid point  $(\Delta, r_p k_p, C_{\text{th}}) = (0.5, 1.0, 0.5)$ , for the Gaussian case ( $f_{\text{NL}} = 0$ , calibrated to  $f_{\text{PBH}} = 1$  at  $A_* = 0.131446$ ) and the two candidate values;  $-35/16$  requires higher  $A$  than  $-35/8$  to reach the same  $f_{\text{PBH}}$ , from `pbh_compaction_fnl.py`. The continuity scan underlying the non-monotonicity discussed in the text is reported in `outputs/pbh_compaction_fnl.json`; the required amplitude-ratio grid (Eq. (11)) is in Table III.

gle horizon mass  $M_H = 10^{20}$  g (matching the first pass, rather than the mass-integrated Eq. 66 of Choudhury *et al.*, an  $O(1)$  width-factor deviation for a narrow peak). Full formalism, the itemized deviations from Choudhury *et al.*’s setup (mass-integration, width-factor, and spectrum-shape approximations), and validation are recorded in `research/track_a3_multichannel/PBH_COMPACTIION_NOTE_2026-09-02.md`.

### C. The lab’s own primordial spectrum: a null result

Sec. VB’s spectrum shape is a lognormal stand-in with a *scanned* amplitude, because Choudhury *et al.*’s spectrum is not reconstructible from their published paper. Removing that dependence, Eq. (11)’s amplitude ratio can instead be evaluated for the spectrum this lab’s own matter-contraction model actually predicts: a near-scale-invariant power law,  $n_s - 1 = 12w/(1 + 3w)$ , anchored to the Planck 2018 amplitude  $A_s = 2.1 \times 10^{-9}$  at  $k_* = 0.05 \text{ Mpc}^{-1}$  and tilted by the small deviation from  $w = 0$  ( $n_s = 0.9649$ ), extrapolated across the 10–15 decades to

coincides with the Planck 2018 reduced Hubble parameter  $h$ , not the dark-matter density parameter ( $\Omega_{\text{DM}} \approx 0.264$ ). Every result in this section is unaffected:  $\Omega_{\text{DM}}$  cancels identically in the amplitude ratio (Eq. (11)) and is absorbed by the Gaussian-case calibration for the tabulated  $f_{\text{PBH}}$  values, so no number here changes with the correct value.

PBH scales under two assumptions: the equation of state does not evolve away from its CMB-calibrated value, and the bounce transfer is scale-independent, which Sec. III A establishes only for  $k\eta_B \lesssim 10^{-2}$ : satisfied at every PBH mass scale considered here only if the bounce energy is high enough. Requiring  $k\eta_B \lesssim 10^{-2}$  at the smallest PBH mass used here ( $10^{15}$  g,  $k \sim 10^{15}\text{--}10^{16} \text{ Mpc}^{-1}$ ) needs  $a_B H_B \gtrsim 10^{17}\text{--}10^{18} \text{ Mpc}^{-1}$ , i.e. a bounce temperature  $T_B \gtrsim 10^8\text{--}10^{10} \text{ GeV}$  – seven decades above the BBN scale used to bound the mass range elsewhere in this section; a scale-independent transfer with  $T_{\text{fNL}} < 1/2$  cannot by itself close the 7-decade amplitude shortfall found below, so this condition, while formally required, does not affect the null verdict. On this shape, the required-amplitude ratio widens from the lognormal grid’s  $1.732 \pm 0.050$  to 1.7–1.9 across a threshold scan  $C_{\text{th}} \in \{0.4, 0.5, 0.6\}$ : the ratio remains the channel’s robust, spectrum-shape-stable output, but  $1.73 \pm 0.05$  should not be quoted as if it were universal.

Fed into the same compaction machinery unmodified, the lab’s own spectrum delivers  $\Delta_\zeta^2 \sim 5 \times 10^{-10}\text{--}1.3 \times 10^{-9}$  at PBH scales against a required 0.006–0.019 (the compaction threshold): a shortfall of 7.0 orders of magnitude in curvature amplitude, at every mass from  $10^{15}$  g to  $10^4 M_\odot$ , at both candidate  $f_{\text{NL}}$  values, over the same threshold scan and a four-decade IR-cutoff scan.  $f_{\text{PBH}}$  is exactly zero in double precision at every point (Gaussian-limit  $\log_{10} \beta \approx -1.7 \times 10^9$  at  $M_H = 10^{20}$  g). **The PBH channel is therefore a null for this model’s own predicted spectrum:** it neither constrains nor is evidenced by primordial black holes, and any quotable PBH signal would require a small-scale amplification mechanism the model does not contain – a literature survey of four such candidate mechanisms [3, 7] finds none applies within the model’s validated ( $k\eta_B \lesssim 10^{-2}$ ) domain.

The same spectrum is checked against the FIRAS  $\mu$ -distortion bound ( $|\mu| < 9 \times 10^{-5}$ ): the lab’s own prediction gives  $\mu = 1.65 \times 10^{-8}$ , safely allowed at  $1.8 \times 10^{-4}$  of the bound. For the early-supermassive-black-hole-seed channel (a possible use of PBH formation this ratio was originally motivated by), the *required* curvature amplitude for  $10^3\text{--}10^6 M_\odot$  seeds is FIRAS-excluded by three orders of magnitude if broadband, and remains excluded for seeds  $\geq 10^4 M_\odot$  even as a narrow ( $\Delta = 0.5$ ) spectral peak; only the lightest ( $10^3 M_\odot$ ) seeds evade the bound with a sufficiently narrow peak. Combined with the null above, the honest statement is two-part: the required SMBH-seed amplitude is largely excluded independent of this model, and this model’s own spectrum falls far short of supplying it either way. Full derivation, the four-mechanism literature survey, and every number above are archived in `research/track_a3_multichannel/inlab_delta2_zeta_2026-09-03.py`.



## VI. CHANNEL III: THE LARGE-SCALE-STRUCTURE BISPECTRUM

### A. Current constraints

The DESI DR1 analysis of 1,631,716 luminous red galaxies ( $0.6 < z < 1.1$ ) and 1,189,129 quasars ( $0.8 < z < 3.1$ ) [18] gives  $f_{\text{NL}}^{\text{local}} = -3.6_{-9.1}^{+9.0}$  (asymmetric 9.0/9.1) at 68% confidence with a merger model for the quasar response, or  $f_{\text{NL}}^{\text{local}} = +3.5_{-7.4}^{+10.7}$  (asymmetric 10.7/7.4) assuming universality. These two numbers come from mutually exclusive theoretical priors on the same DESI DR1 data (merger-response vs. universality) and are not directly comparable to each other; each is quoted here against its own prior. The prediction of Eq. (1) sits  $0.16\sigma$  from the former and  $0.77\sigma$  from the latter. The discriminating power is  $|f_{\text{NL}}|/\sigma = 0.24$ : DESI DR1 is consistent with the prediction under either prior and cannot test it.

### B. Forecast reach

As established in Sec. III, the scheme-S1  $k\eta_B$  window is satisfied most easily at this channel's CMB/LSS pivot scale, so the transmitted amplitude  $f_{\text{NL}}^{\text{after}} = T_{f_{\text{NL}}} \cdot f_{\text{NL}}^{\text{pre}} + \Delta f_{\text{NL}}^{\text{bounce}} \in [-0.65, -0.50]$  ( $-35/16$ ) or  $[-1.20, -0.86]$  ( $-35/8$ ) (Sec. III A, Table in that section) is this paper's observable prediction for Channel III, not the pre-bounce contraction-phase amplitude. Table IV reports the bare significance  $f_{\text{NL}}^{\text{after}}/\sigma_{f_{\text{NL}}}$  across the three backgrounds of Sec. III, with the pre-bounce amplitude's bare significance retained as a clearly-labelled secondary row (i.e. assuming  $T_{f_{\text{NL}}} = 1$ , the transfer-free case this paper does not otherwise support). The matter-bounce bispectrum is also not exactly the local shape; a noise-weighted shape overlap  $r < 1$ , reported in this lab's companion Fisher-forecast draft (referenced in the reproducibility statement below) but not re-derived at the  $f_{\text{NL}}^{\text{after}}$  fiducial used in this paper – so no numeral is quoted here – would let an experiment quoting  $\sigma_{\text{local}}$  constrain the bounce amplitude at  $\sigma_{\text{local}}/r$ ; the  $r$ -projected numbers are omitted pending that re-derivation.

At the transmitted amplitude the two candidate values separate by  $|\Delta f_{\text{NL}}^{\text{after}}| \approx 0.35\text{--}0.55$  against  $\sigma_{f_{\text{NL}}} = 0.5\text{--}0.7$  at SPHEREx sensitivity: under  $1\sigma$ , so this channel does not discriminate  $-35/16$  from  $-35/8$  at current forecast reach once scheme-S1 transmission is applied; it remains the only channel where the factor of two would be observationally consequential *if* the pre-bounce amplitude survived transmission unsuppressed, which Sec. III shows it does not.

TABLE IV. Channel III. Bare detection significance  $|f_{\text{NL}}^{\text{after}}|/\sigma_{f_{\text{NL}}}$  of the scheme-S1 transmitted amplitude across the three backgrounds of Sec. III (range shown; Table in Sec. III A gives per-background values). The pre-bounce row (assuming  $T_{f_{\text{NL}}} = 1$ ) is a secondary reference, not this paper's prediction. Forecast  $\sigma$  values are quoted from the cited abstracts. MegaMapper is an unapproved design and is quoted only at its published order-unity level; no systematic budget is transferred to it. Values re-derived by `survey_reach_fnl.py` (`outputs/survey_reach_fnl.json`).

| Survey                                                                     | $\sigma_{f_{\text{NL}}}$ | bare ( $-35/16$ )         | bare ( $-35/8$ )          |
|----------------------------------------------------------------------------|--------------------------|---------------------------|---------------------------|
| SPHEREx, bispectrum only [19]                                              | 0.7                      | $0.7\text{--}0.9\sigma$   | $1.2\text{--}1.7\sigma$   |
| SPHEREx, $P + B$ target [19, 20]                                           | 0.5                      | $1.0\text{--}1.3\sigma$   | $1.7\text{--}2.4\sigma$   |
| MegaMapper class [21, 22]                                                  | $\sim 1$                 | $0.50\text{--}0.65\sigma$ | $0.86\text{--}1.20\sigma$ |
| <i>pre-bounce secondary (assuming <math>T_{f_{\text{NL}}} = 1</math>):</i> |                          |                           |                           |
| SPHEREx, bispectrum only                                                   | 0.7                      | $3.13\sigma$              | $6.25\sigma$              |
| SPHEREx, $P + B$ target                                                    | 0.5                      | $4.38\sigma$              | $8.75\sigma$              |
| MegaMapper class                                                           | $\sim 1$                 | $2.19\sigma$              | $4.38\sigma$              |

## VII. DISCUSSION

### A. The joint statement

The matter-bounce prediction  $f_{\text{NL}}^{\text{local}} = -35/16$  – independently re-derived here within the in-in formalism, confirming Li *et al.* (2017) and localizing the factor-of-two slip in Cai *et al.* (2009), and now confirmed by a method-independent route (Sec. II D) – is consistent with every channel currently able to speak. Pulsar timing is consistent with the associated induced-GW slope  $\gamma = 3$  under both NANOGrav's official posterior and this lab's free-spectrum refit, but is insensitive to the value of  $f_{\text{NL}}$  and therefore does not test it. Primordial black holes yield a reproducible amplitude ratio ( $1.7\text{--}1.9$ ) between the two candidate values, but the lab's own predicted spectrum falls 7.0 orders of magnitude short of the required amplitude at every mass scale considered (Sec. V C): the channel is a clean null for this model, not an open promise. Transmission through the bounce is now a definite, scheme-qualified prediction rather than an open bound: within scheme S1 and  $k\eta_B \lesssim 10^{-2}$ ,  $f_{\text{NL}}^{\text{after}} \in [-0.65, -0.50]$  (Sec. III A). As Sec. III states, this  $k\eta_B$  window is an upper bound on  $k$  and is satisfied *most easily*, not *hardest*, at the CMB/LSS pivot scale used by Channel III below, so this transmitted range is precisely the number that channel should be compared against. Galaxy surveys are the only channel that discriminates between the two candidate amplitudes at all, but at the transmitted amplitude that discrimination is weak: DESI DR1 is consistent with  $f_{\text{NL}}^{\text{after}}$  at well under  $1\sigma$  under either prior, and SPHEREx reaches only  $0.7\text{--}0.9\sigma$  ( $-35/16$ ) or  $1.2\text{--}1.7\sigma$  ( $-35/8$ ) bare significance (Table IV) – under  $1\sigma$  separation between the two candidates at current or forecast LSS sensitivity, not the  $3.13\sigma$  apparent tension with the pre-bounce amplitude a naive reading of Channel III alone would suggest. No channel

is in tension with another.

## B. Where the consistency is weak

The pulsar-timing result is a statement about a slope ( $\gamma = 3$ ) that, as Sec. IV details, requires the matter bounce’s specific broad, near-scale-invariant scalar source rather than following from causality alone – a generic causal floor would instead give  $\gamma = 2$  ( $\Omega_{\text{GW}} \propto f^3$ ) [9] – but it is identical at both candidate  $f_{\text{NL}}$  values regardless. The black hole channel now uses the literature’s own formation criterion and the lab’s own spectrum rather than an external, unreconstructible one, and returns a definite verdict (null) rather than an open amplitude scan; the amplitude ratio remains the channel’s only shape-robust, spectrum-independent output, and does not by itself separate  $-35/16$  from  $-35/8$  observationally – only compares how hard each is to produce – and  $f_{\text{PBH}}(f_{\text{NL}})$  itself was shown to be strongly non-monotonic along the path to both candidates on the earlier lognormal grid (Sec. VB). The transmission result is scheme-conditional (S1) and validated only for  $k\eta_B \lesssim 10^{-2}$ ; a second scheme (S2) does not regulate at all, so scheme-independence remains open. At the transmitted amplitude the survey channel’s discriminating power is itself the weak link: SPHEREx separates the two candidates by  $|\Delta f_{\text{NL}}^{\text{after}}| \approx 0.35\text{--}0.55$  against  $\sigma_{f_{\text{NL}}} = 0.5\text{--}0.7$ , i.e. under  $1\sigma$  (Sec. VI), so no currently forecast LSS survey can distinguish  $-35/16$  from  $-35/8$  once scheme-S1 transmission is applied; the not-yet-derived shape-overlap projection  $r$  would sharpen this somewhat but cannot by itself restore the pre-bounce-level  $3\sigma$  separation.

## C. Next steps

In priority order: (i) a second transmission scheme beyond S1, to test whether  $f_{\text{NL}}^{\text{after}} \in [-0.65, -0.50]$  is scheme-independent, and extension of the bounce-cubic-term computation to a regulated S2 value; (ii) propagation of the scheme-S1 post-bounce amplitude to the CMB/LSS pivot scale used by Channel III, which requires establishing the transmission result’s  $k\eta_B$  validity at that far larger scale rather than assuming it; (iii) propagation of the matter-bounce scalar spectrum through the induced-GW kernel to predict the nHz *amplitude*, converting Channel I from a consistency check into a test; (iv) re-derivation of the shape overlap  $r$  at the  $-35/16$  fiducial; and (v) a direct-collapse (rather than horizon-re-entry) primordial-black-hole formation channel during the dust contraction phase itself, identified but not evaluated in Sec. VC.

## Appendix A: The $\delta N_c$ – $\zeta_{\text{Mald}}$ derivation

This appendix transcribes, without new science, the derivation archived in `research/theory_audit/fnl_monopole_adjudication_2026_09_03.md` (§§1–4) that underlies Eq. (5) and the  $[L]/[K]/[X]/[S]$  decomposition of Sec. IID.

### 1. The linear-order identity, Eq. (5)

On a uniform- $\phi$  slice the lapse is  $N = 1 + \dot{\zeta}/H$ , so the Friedmann equation at fixed  $\phi$  gives  $\delta H/H = -(\epsilon/3)\dot{\zeta}_L/H$ : the local expansion rate on comoving slices is perturbed at  $O(k^0)$ , carried by the trace of the shift gradient,  $\delta K = -\partial^2\psi/a^2 \rightarrow -\epsilon\dot{\zeta}_L$ . The separate-universe  $\delta N$  is defined on the zero-shift threading,  $\delta N_c = \int (K/3) d\tau$ , so writing  $\delta N_c = \zeta + b$  with  $\dot{b} = \delta K/3$  gives, on the growing mode,  $b = -(\epsilon/3)\zeta_L$ , i.e. Eq. (5),  $\delta N_c = (1 - \epsilon/3)\zeta_{\text{Mald}}$  at linear order (sign convention:  $\zeta_{\text{Mald}}$  and  $\delta N_c$  share the sign of the comoving-gauge curvature perturbation used throughout this paper).

### 2. The $[L]/[K]/[X]/[S]$ decomposition

Separating the long mode’s coupling to the short-mode pair by how it enters:

| class                             | long mode enters as                                              | squeezed value at $\epsilon = 3/2$                          | value at $\epsilon = 3/2$ |
|-----------------------------------|------------------------------------------------------------------|-------------------------------------------------------------|---------------------------|
| $[L]$                             | $\zeta_L, \dot{\zeta}_L$<br>(lapse/curvature;<br>local response) | $-25/8$ ,<br>monopole                                       | pure                      |
| $[K]$                             | $\partial_i \partial_j \tilde{\chi}_L$ ( $\delta K$ and shear)   | 0 (cancels between<br>varied legs of $T_4$ )                |                           |
| $[X]$                             | $\partial_i \tilde{\chi}_L$ (the<br>$\propto 1/k_L$ )            | $\frac{15}{16}$ + $\frac{15}{16}\mu^2$<br>(monopole $5/4$ ) |                           |
| $[S]$                             | short-mode sourcing of<br>the long mode                          | 0 (suppressed)                                              |                           |
| total                             |                                                                  | $-\frac{35}{16} + \frac{15}{16}\mu^2$ (= in-<br>in exactly) |                           |
| comoving $\delta N$ (lanes A/B/C) | different variable, no                                           | $-5$ , isotropic                                            |                           |
|                                   | $[X]$                                                            |                                                             |                           |

The  $1/k_L$  poles in  $[X]$ ’s two contributions – the  $T_3$  bulk coupling and the boundary term  $f_b$  below – cancel only in their sum, leaving a finite  $O(k_L^0)$  remainder; no separate-universe construction (isotropic or Bianchi-I) contains this term, because it originates in the comoving-gauge *shift*  $N_i = \partial_i\psi \supset a^2\epsilon\partial^{-2}\zeta$ , which is  $O(1/k_L)$  in this non-attractor phase and so violates the gradient-expansion assumption  $N_i = O(\nabla)$  every separate-universe construction relies on. The final-time re-threading boundary term,

$$f_b = \frac{\epsilon}{2\mathcal{H}} \left[ \partial\zeta \partial\tilde{\chi} - \partial^{-2}\partial_i\partial_j(\partial_i\zeta \partial_j\tilde{\chi}) \right], \quad (\text{A1})$$

subtracts  $(1 - \epsilon/3) \times \frac{5\epsilon}{4}\mu^2$  from the  $T_3$  bulk’s  $\mu^2$  coupling and adds the  $+15/16$  monopole, reproducing the

quadrupole exactly and supplying the  $[X]$  monopole. In-in monopole minus  $\delta N_c$ :  $-\frac{15}{8} - (-5) = \frac{25}{8} = [X]_{\text{mono}}(5/4) + ([L] - \delta N_c)(15/8)$ ; in general  $\epsilon$ ,  $5\epsilon(9 - \epsilon)/18 = 5\epsilon(9 - 2\epsilon)/36 + 5\epsilon/4$ . The second piece is the second-order part of the  $\zeta_{\text{Mald}} \rightarrow \delta N_c$  threading map; it is *recorded* as exactly the pure pair-translation coefficient  $(5/12)(3\epsilon) -$  a computed identity, not a claimed mechanism.

### 3. General- $\epsilon$ formulas

The comoving separate-universe results, for general  $\epsilon$ , are

$$f_{\text{NL}}^p = \frac{5(\epsilon - 7)}{8}, \quad f_{\text{NL}}^c = -5 \quad (\text{all } \epsilon), \quad (\text{A2})$$

and the in-in squeezed bispectrum, general  $\epsilon$ , is

$$f^{\text{sq}}(\mu, \epsilon) = \frac{5}{12}(\epsilon^2 \mu^2 - \epsilon^2 + 6\epsilon - 12), \quad (\text{A3})$$

with monopole  $-5(\epsilon - 3)(\epsilon - 6)/18$ , isosceles ( $\mu = 0$ ) value  $-5(\epsilon^2 - 6\epsilon + 12)/12$ , and  $\mu^2$  coefficient  $5\epsilon^2/12$ ; at  $\epsilon = 3/2$  these reduce to the monopole  $-15/8$ , the isosceles value  $-35/16$  of Eq. (4)’s  $\mu = 0$  limit, and the  $+15/16 \mu^2$  quadrupole quoted throughout this paper.

### 4. Bianchi-I: why an anisotropic separate universe does not close the gap

An anisotropic (Bianchi-I) separate-universe treatment, carrying the long mode’s projected shear onto the short-mode power spectrum exactly, gives an identically zero monopole contribution: the shear is  $\mathcal{O}(k_L^0)$  and traceless, and no traceless linear-in- $\zeta_L$  response can supply a monopole. It therefore cannot by itself close the  $-25/8$  gap between  $\delta N_c = -5$  and the in-in monopole  $-15/8$ ; the gap is closed only by  $[X]$ , the shift term no separate-universe construction contains (Sec. A2 above).

## REPRODUCIBILITY STATEMENT

Every result reported here traces to a committed script, chain, or reproducibility manifest; nothing is quoted from memory. The exact-amplitude derivation (Sec. II) and the bounce-transmission bound (Sec. III) are committed, pinned to commit 68309c8, at

[https://github.com/Hubify-Projects/bigbounce/tree/68309c8/research/theory\\_audit](https://github.com/Hubify-Projects/bigbounce/tree/68309c8/research/theory_audit)

[https://github.com/Hubify-Projects/bigbounce/tree/68309c8/research/cubic\\_bounce\\_transmission](https://github.com/Hubify-Projects/bigbounce/tree/68309c8/research/cubic_bounce_transmission)

respectively. The bounce-to-local shape overlap ( $r$ , Sec. VI) is imported from this lab’s companion Fisher-forecast draft at [research/focused\\_paper\\_source\\_integration/](#) and [outputs/survey\\_reach\\_fnl.json](#). The three

channel computations of Secs. IV–VI are committed at [research/track\\_a3\\_multichannel/pta\\_gamma\\_reproduce.py](#), [pbh\\_abundance\\_fnl.py](#) (first-pass Press-Schechter, retained for comparison only), [pbh\\_compaction\\_fnl.py](#) (compaction-function criterion, this paper’s principal PBH result), [inlab\\_delta2\\_zeta\\_2026-09-03.py](#) (Sec. VC, the lab’s own predicted spectrum and its FIRAS check), [survey\\_reach\\_fnl.py](#), and [pta\\_injection\\_30bin\\_2026-09-02.py](#) (the Sec. IVC injection-recovery validation), with results in [outputs/\\*.json](#) and manifests at [reproducibility/manifests/experiments/a3-\\*.json](#) (PTA/PBH/reach, including [a3-pbh-compaction-fnl.json](#), [reproducibility/manifests/experiments/p2-fnl-\\*.json](#) (exact-amplitude derivation and validation), and [reproducibility/manifests/experiments/p2-a2-\\*.json](#) (bounce-transmission computation). The pulsar-timing chain is [pipelines/p3\\_pta\\_mcmc/free\\_spectrum\\_real\\_2026-05-01/chain\\_real\\_freespec.npy](#) (SHA-256, in two halves to allow line-breaking: [50abc38a04e1bf886adc833b5eb653be4e986877fe2476f87a78927f4f3610fc](#)), derived from the NANOGrav 15-yr KDE free spectra, Zenodo [10.5281/zenodo.8060824](#). This work’s own code and artifacts are pinned by the GitHub commit hash above rather than a frozen-release DOI of their own; minting one is a maintainer action planned for the packaging stage. The exact-amplitude theory lineage this paper builds on is archived at this lab’s companion P2 Zenodo record, [10.5281/zenodo.21461881](#). Every computation reported in this paper ran on a local CPU (Apple M5, 24 GB, macOS 26.5) at zero compute cost; the PTA/PBH/reach/injection channels ran in under 6 s total. Compute venue, wall-clock time, and external data sources are recorded per-experiment in the manifests cited above, consistent with this lab’s standing reproducibility policy (directive Q2).

## AI USAGE DISCLOSURE

Symbolic derivations reported in Secs. II–III were performed by deterministic sympy scripts written and executed under the author’s direction; every intermediate result is printed by the committed scripts and independently checked against the cited literature limits before use. The channel computations of Secs. IV–VI were performed by committed Python scripts against real data and cited published values, not by an LLM. Claude (Anthropic) assisted with literature cross-checks, prose drafting, and L<sup>A</sup>T<sub>E</sub>X formatting under the author’s direction; all physics content, equations, and numerical results were verified by the author against the committed script output.

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- [1] Y.-F. Cai, W. Xue, R. Brandenberger, and X. Zhang, “Non-Gaussianity in a Matter Bounce,” JCAP **0905**, 011 (2009), arXiv:0903.0631.
  - [2] Y.-F. Cai, D. A. Easson, and R. Brandenberger, “Towards a Nonsingular Bouncing Cosmology,” JCAP **1208**, 020 (2012), arXiv:1206.2382.
  - [3] J. Quintin, Z. Sherkatghanad, Y.-F. Cai, and R. H. Brandenberger, “Evolution of cosmological perturbations and the production of non-Gaussianities through a nonsingular bounce: Indications for a no-go theorem in single field matter bounce cosmologies,” Phys. Rev. D **92**, 063532 (2015), arXiv:1508.04141.
  - [4] Y.-B. Li, J. Quintin, D.-G. Wang, and Y.-F. Cai, “Matter bounce cosmology with a generalized single field: non-Gaussianity and an extended no-go theorem,” JCAP **1703**, 031 (2017), arXiv:1612.02036.
  - [5] J. Maldacena, “Non-Gaussian features of primordial fluctuations in single field inflationary models,” JHEP **05**, 013 (2003), arXiv:astro-ph/0210603.
  - [6] M. H. Namjoo, H. Firouzjahi, and M. Sasaki, “Violation of non-Gaussianity consistency relation in a single field inflationary model,” Europhys. Lett. **101**, 39001 (2013), arXiv:1210.3692.
  - [7] I. Agullo, B. Bolliet, and V. Sreenath, “Non-Gaussianity in loop quantum cosmology,” Phys. Rev. D **97**, 066021 (2018), arXiv:1712.08148.
  - [8] T. Papanikolaou, “Gravitational wave signatures of non-singular matter bouncing cosmology in NANOGrav and beyond,” arXiv:2504.11641.
  - [9] Y. Cai, S. Pi, and M. Sasaki, “Gravitational Waves Induced by non-Gaussian Scalar Perturbations,” Phys. Rev. D **102**, 083528 (2020), arXiv:1909.13728.
  - [10] G. Agazie *et al.* (NANOGrav Collaboration), “The NANOGrav 15 yr Data Set: Evidence for a Gravitational-wave Background,” Astrophys. J. Lett. **951**, L8 (2023), arXiv:2306.16213.
  - [11] NANOGrav Collaboration, “NANOGrav 15 yr KDE Free Spectra v1.0.0,” Zenodo, [10.5281/zenodo.8060824](https://doi.org/10.5281/zenodo.8060824).
  - [12] S. Young and C. T. Byrnes, “Primordial black holes in non-Gaussian regimes,” JCAP **1308**, 052 (2013), arXiv:1307.4995.
  - [13] G. Franciolini, A. Kehagias, S. Matarrese, and A. Riotto, “Primordial Black Holes from Inflation and non-Gaussianity,” JCAP **1803**, 016 (2018), arXiv:1801.09415.
  - [14] M. Sasaki, T. Suyama, T. Tanaka, and S. Yokoyama, “Primordial black holes – perspectives in gravitational wave astronomy,” Class. Quant. Grav. **35**, 063001 (2018), arXiv:1801.05235.
  - [15] S. Choudhury, K. Dey, S. Ganguly, A. Karde, S. K. Singh, and P. Tiwari, “Negative non-Gaussianity as a salvager for PBHs with PTAs in bounce,” Eur. Phys. J. C **85**, 472 (2025), arXiv:2409.18983.
  - [16] S. Young, C. T. Byrnes, and M. Sasaki, “Calculating the mass fraction of primordial black holes,” JCAP **07**, 045 (2019), arXiv:1904.00984.
  - [17] I. Musco, “Threshold for primordial black holes: Dependence on the shape of the cosmological perturbations,” Phys. Rev. D **100**, 123524 (2019), arXiv:1809.02127.
  - [18] E. Chaussidon *et al.* (DESI Collaboration), “Constraining primordial non-Gaussianity with DESI 2024 LRG and QSO samples,” arXiv:2411.17623.
  - [19] C. Heinrich, O. Doré, and E. Krause, “Measuring  $f_{\text{NL}}$  with the SPHEREx Multi-tracer Redshift Space Bispectrum,” arXiv:2311.13082.
  - [20] O. Doré *et al.*, “Cosmology with the SPHEREx All-Sky Spectral Survey,” arXiv:1412.4872.
  - [21] S. Ferraro *et al.*, “Inflation and Dark Energy from spectroscopy at  $z > 2$ ,” Astro2020 science white paper, arXiv:1903.09208.
  - [22] N. Sailer, E. Castorina, S. Ferraro, and M. White, “Cosmology at high redshift – a probe of fundamental physics,” JCAP **12**, 049 (2021), arXiv:2106.09713.